

VCS MODULE VMD0005

REDD METHODOLOGICAL MODULE:

ESTIMATION OF CARBON STOCKS IN THE
LONG-TERM WOOD PRODUCTS POOL
(CP-W)

Version 1.1

20 November 2012

Sectoral Scope 14

Methodology developed by:



Revision prepared by Winrock International and TerraCarbon

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1 SOURCES

This module is one of numerous modules that comprise the VCS approved methodology VM0007: REDD Methodology Modules.

2 SUMMARY DESCRIPTION OF THE MODULE

This module allows for *ex ante* estimation of carbon stocks in the long-term wood products pool in the baseline case. Carbon stocks treated here are those stocks entering the wood products pool at the time of deforestation.

3 APPLICABILITY CONDITIONS

This module is applicable to all cases where wood is harvested for conversion to wood products *for commercial markets*, for all forest types and age classes. This module is applicable in the baseline if the wood products pool is included as part of the project boundary as per applicability criteria in the framework module **REDD-MF**, specifically:

- Timber harvest occurs prior to or in the process of deforestation, and where timber is destined for commercial markets
- The wood products pool is determined to be significant (using **T-SIG**).

4 PROCEDURES

Parameters

This module produces the following parameter:

Parameter	SI Unit	Description
$C_{WP,i}$	t CO ₂ -e ha ⁻¹	Mean carbon stock entering the wood products pool at the time of deforestation from stratum <i>i</i>
$C_{XB,i}$	t CO ₂ -e ha ⁻¹	Mean stock of extracted biomass carbon from stratum <i>i</i>

Frequency of update of oxidation factors

The approach outlined in this module employs an emission factor (WW) derived by Winjum *et al.* 1998. In the event that new research findings updating or refining (e.g. for specific countries) the WW factor become available in the future (during the project crediting period), they will replace the factors included in the module, otherwise the factors in the module will remain valid. The use of this module requires that project proponents review research findings (that produce emissions factors compatible with the conceptual framework here) at least every 10 years to identify further refinements to the emission factors that are empirically-based and peer-reviewed.

Ex ante estimation of carbon stocks in the wood products pool in the baseline

This module estimates carbon stocks in wood products resulting from timber harvest occurring prior to or in the process of deforestation. Accounting for such wood products must only take place at the time of deforestation.

All factors are derived from Winjum *et al.* 1998.

If approved timber harvest plans, specifying harvest intensity per strata in terms of volume extracted per ha, are available for the project area use Option 1. If approved harvest plans are not available use Option 2.

Option 1: Direct Volume Extraction Estimation

Step 1: Identify the wood product class(es) (*ty*; defined here as sawnwood, wood-based panels, other industrial roundwood, paper and paper board, and other) that are the anticipated end use of the extracted carbon calculated in Step 2.

Step 2: Calculate the biomass carbon of the volume extracted by wood product type *ty* from within the project boundary:

$$C_{XB,ty,i} = \frac{1}{A_i} * \sum_{j=1}^S (V_{ex,ty,j,i} * D_j * CF_j * \frac{44}{12}) \quad (1)$$

Where:

$C_{XB,ty,i}$	Mean stock of extracted biomass carbon by class of wood product <i>ty</i> from stratum <i>i</i> ; t CO ₂ -e ha ⁻¹
A_i	Total area of stratum <i>i</i> ; ha
$V_{ex,ty,j}$	Volume of timber extracted from within stratum <i>i</i> (does not include slash left onsite) by species <i>j</i> and wood product class <i>ty</i> ; m ³
D_j	Mean wood density of species <i>j</i> ; t d.m.m ⁻³
CF_j	Carbon fraction of biomass for tree species <i>j</i> ; t C t ⁻¹ d.m.
<i>j</i>	1, 2, 3, ... S tree species
<i>ty</i>	Wood product class – defined here as sawnwood (s), wood-based panels (w), other industrial roundwood (oir), paper and paper board (p), and other (o)
44/12	Ratio of molecular weight of CO ₂ to carbon, t CO ₂ -e t C ⁻¹

Step 3: Calculate the biomass carbon extracted that enters the wood products pool at the time of deforestation.

$$C_{WP,i} = \sum_{ty=s,w,oir,p,o} C_{XB,ty,i} * (1 - WW_{ty}) \quad (2)$$

Where:

$C_{WP,i}$	Carbon stock entering the wood products pool from stratum i ; t CO ₂ -e ha ⁻¹
$C_{XB,ty,i}$	Mean stock of extracted biomass carbon by class of wood product ty from stratum i ; t CO ₂ -e ha ⁻¹
WW_{ty}	Wood waste. The fraction immediately emitted through mill inefficiency by class of wood product ty ; dimensionless
ty	Wood product class – defined here as sawnwood (s), wood-based panels (w), other industrial roundwood (oir), paper and paper board (p), and other (o)
i	1, 2, 3, ... M strata

Step 4: Calculate the amount of wood products entering the pool at the time of deforestation ($C_{WP,i}$ calculated in C-WP) that is expected to be emitted over a 100-year timeframe.

$$C_{WP100,i} = C_{WP,i} - C_{WP,i} * (1 - SLF_p) * (1 - OF_p) \quad (3)$$

Where,

$C_{WP100,i}$	Carbon stock entering the wood products pool at the time of deforestation that is expected to be emitted over 100-years from stratum i ; t CO ₂ -e ha ⁻¹
$C_{WP,i}$	Carbon stock entering wood products pool at time of deforestation from stratum i ; t CO ₂ -e ha ⁻¹
SLF_{ty}	Fraction of wood products that will be emitted to the atmosphere within 5 years of timber harvest by class of wood product ty ; dimensionless
OF_{ty}	Fraction of wood products that will be emitted to the atmosphere between 5 and 100 years of timber harvest by class of wood product ty ; dimensionless
ty	Wood product class – defined here as sawnwood (s), wood-based panels (w), other industrial roundwood (oir), paper and paper board (p), and other (o)
i	1, 2, 3, ... M strata

Option 2: Commercial inventory estimation

Step 1: Calculate the biomass carbon of the commercial volume extracted prior to or in the process of deforestation:

$$C_{XB,i} = C_{AB_tree,i} * \frac{1}{BCEF} * Pcom_i \quad (4)$$

Where:

$C_{XB,i}$	Mean stock of extracted biomass carbon from stratum i ; t CO ₂ -e ha ⁻¹
$C_{AB_tree,i}$	Mean aboveground biomass carbon stock in stratum i ; t CO ₂ -e ha ⁻¹
$BCEF$	Biomass conversion and expansion factor (BCEF) for conversion of merchantable volume to total aboveground tree biomass; dimensionless
$Pcom_i$	Commercial volume as a percent of total aboveground volume in stratum i ; dimensionless
i	1, 2, 3, ... M strata

Step 2: Identify the wood product class(es) (ty ; defined here as sawnwood, wood-based panels, other industrial roundwood, paper and paper board, and other) that are the anticipated end use of the extracted carbon calculated in Step 1.

Step 3: Calculate the biomass carbon entering the wood products pool at the time of deforestation.

$$C_{WP,i} = \sum_{ty=s,w,oir,p,o} C_{XB,ty,i} * (1 - WW_{ty}) \quad (5)$$

Where:

$C_{WP,i}$	Carbon stock entering the wood products pool from stratum i ; t CO ₂ -e ha ⁻¹
$C_{XB,ty,i}$	Mean stock of extracted biomass carbon by class of wood product ty from stratum i ; t CO ₂ -e ha ⁻¹
WW_{ty}	Wood waste. The fraction immediately emitted through mill inefficiency by class of wood product ty ; dimensionless
ty	Wood product class – defined here as sawnwood (s), wood-based panels (w), other industrial roundwood (oir), paper and paper board (p), and other (o)
i	1, 2, 3, ... M strata

Step 4: Calculate the amount of wood products entering the pool at the time of deforestation ($C_{WP,i}$, calculated in C-WP) that is expected to be emitted over a 100-year timeframe.

$$C_{WP100,i} = C_{WP,i} - C_{WP,i} * (1 - SLF_p) * (1 - OF_p) \quad (6)$$

Where,

$C_{WP100,i}$	Carbon stock entering the wood products pool at the time of deforestation that is expected to be emitted over 100-years from stratum i ; t CO ₂ -e ha ⁻¹
$C_{WP,i}$	Carbon stock entering wood products pool at time of deforestation from stratum i ; t CO ₂ -e ha ⁻¹
SLF_{ty}	Fraction of wood products that will be emitted to the atmosphere within 5 years of timber harvest by class of wood product ty ; dimensionless
OF_{ty}	Fraction of wood products that will be emitted to the atmosphere between 5 and 100 years of timber harvest by class of wood product ty ; dimensionless
ty	Wood product class – defined here as sawnwood (s), wood-based panels (w), other industrial roundwood (oir), paper and paper board (p), and other (o)
i	1, 2, 3, ... M strata

5 PARAMETERS

5.1. Data and Parameters Not Monitored

Data Unit / Parameter:	BCEF
Data unit:	Dimensionless
Used in equations:	1
Description:	Biomass conversion and expansion factor for conversion of commercial wood volume per unit area to total aboveground tree biomass per unit area; note that BCEF as defined here, and in most applications, is not applied on a per stem basis

Source of data:	Equations have been derived using a wide range of measured variables (commercial wood volume per unit area and total aboveground biomass per unit area) based on datasets that comprise at least 30 trees. Equations must be based on statistically significant regressions and have an r^2 that is ≥ 0.8. The source of data shall be chosen with priority from higher to lower preference as follows: (a) Existing local forest type-specific; (b) National forest type-specific or eco-region-specific (e.g. from national GHG inventory); (c) Forest type-specific or eco-region-specific from neighboring countries with similar conditions. Sometimes (c) might be preferable to (b); (d) Global forest type or eco-region-specific (e.g. IPCC 2006 INV GLs AFOLU Chapter 4 Table 4.5) The project volume data to which the selected BCEF is applied must conform to the data the BCEF was originally derived from, in particular, it must match forest type, stand structure, minimum DBH, and cover the range of potential independent variable values (commercial volumes) likely to be encountered in the project area. Care must be taken to ensure that the selected BCEF does not account for non-commercial species not represented in commercial volume estimates (i.e. is restricted to expanding merchantable volumes to account for only non-merchantable tree components).
Justification of choice of data or description of measurement methods and procedures applied:	-
Any comment:	Alternatively, BCEF, where not directly available, can be calculated as wood density (t dry mass m^{-3} green volume) * BEF (Biomass Expansion Factor = ratio of aboveground biomass to biomass of the commercial volume). If using BCEFs developed outside the project country (cases (c) and (d) above under Source of data), it is necessary to validate the applicability of BCEFs used. Validation is performed by:

	1. Limited Measurements • Select at least 20 plots in the project area covering a wide range of commercial volumes. • Obtain tree measurements (e.g. DBH, height to a 10 cm diameter top) from which to calculate commercial volume and total biomass. • Calculate commercial volume per unit area (e.g. using Smalian's formula) and total biomass per unit area (using the biomass equation(s) selected for application in CP-AB) for each plot • Calculate BCEF for each plot (biomass (t) / commercial volume (m³)) Graph the plot-level estimates of BCEF versus commercial volume along with the BCEF equation (predicted) to be validated. If the estimated BCEFs of the measured plots are distributed both above and below the predicted value the BCEF equation may be used. The BCEF equation may also be used if the measured plots have a BCEF consistently lower than that predicted. If graphing the BCEF of the measured plots indicates a systematic bias to overestimation of BCEF (>75% of the plots below the predicted value) then another BCEF equation must be selected or developed anew.
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Data Unit / Parameter:	CF_j
Data unit:	t C t ⁻¹ d.m.
Used in equations:	4
Description:	Carbon fraction of dry matter in t C t ⁻¹ d.m. for species j
Source of data:	Species- or family-specific values from the literature (e.g. IPCC 2006 INV GLs AFOLU Chapter 4 Table 4.3) shall be used if available, otherwise default value of 0.47 t C t ⁻¹ d.m. can be used.
Justification of choice of data or description of measurement methods and procedures applied:	-
Any comment:	Where new species are encountered in the course of monitoring, new carbon fraction values must be sourced from the literature or otherwise use the default value.

Data Unit / Parameter:	D_j
Data unit:	t d.m. m ⁻³
Used in equations:	4
Description:	Basic wood density in t d.m. m ⁻³ for species j
Source of data:	The source of data shall be chosen with priority from higher to lower preference as follows: (a) National species-specific or group of species-specific (e.g. from National GHG inventory); (b) Species-specific or group of species-specific from neighboring countries with similar conditions. Sometimes (b) may be preferable to (a); (c) Global species-specific or group of species-specific (e.g. IPCC 2006 INV GLs AFOLU Chapter 4 Tables 4.13 and 4.14). Species-specific wood densities may not always be available, and may be difficult to apply with certainty in the typically species rich forests of the humid tropics, hence it is acceptable practice to use wood densities developed for forest types or plant families or species groups.
Justification of choice of data or description of measurement methods and procedures applied:	-
Any comment:	Where using wood densities developed outside of the project country (cases (b) and (c) above under Source of data), wood densities must be validated with either limited destructive sampling or direct measurement of wood hardness (e.g. with a Pilodyn wood tester) in the field and correlating with wood density. Samples or measurements must be from 20-30 trees. For validation of mean forest type or species group wood densities, representation of species in the sample must be proportional to their occurrence in terms of basal area or volume in the project area (not abundance or stem density). Samples must provide representation across the length of the tree. Wood samples are cut in discs and thickness and diameter measured to calculate green volume. Samples are oven dried (70° C) to a constant weight in the laboratory, and density calculated as dry weight (g) per unit green volume (cm³).

	If the density of the samples/measurements (or mean density in the case of forest type or species group means) is within $\pm 10\%$ of the selected density values, then the selected density values may be used. Otherwise, a new density value must be developed with more extensive sampling, using the validation samples as a base. Where new species are encountered in the course of monitoring, new wood density values must be sourced from the literature and validated, if necessary, as per requirements and procedures above.
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Data Unit / Parameter:	$Pcom_i$
Data unit:	Dimensionless
Used in equations:	1
Description:	Commercial volume as a percent of total aboveground volume in stratum i .
Source of data:	The source of data shall be chosen with priority from higher to lower preference as follows: (a) Direct forest inventory of the project area, distinguishing commercially viable stocks on the basis of species and tree size, referencing local expert knowledge or a participatory rural assessment (PRA) of harvest practices and markets; (b) Forest inventory from a proxy area in the same region, representing the same forest type and age class, distinguishing commercially viable stocks on the basis of species and tree size, referencing local expert knowledge of harvest practices and markets National and forest type-specific or eco-region-specific (e.g. from National GHG inventory).
Justification of choice of data or description of measurement methods and procedures applied:	This parameter is updated at baseline renewal when aboveground biomass is re-inventoried as per module CP-AB (at least every 10 years).
Any comment:	Updated at the time of baseline revision (at least every 10 years). Note that application of the commercial percentage of total volume introduces the simplifying assumption (and conservative, as it is only used in the <i>ex-ante</i> baseline calculations) that all commercial stocks are extracted (i.e. perfect efficiency).

Data Unit / Parameter:	SLF_{ty}																				
Data unit:	Dimensionless																				
Used in equations:	2, 4																				
Description:	SLF = Fraction of wood products that will be emitted to the atmosphere within 5 years of production by class of wood product ty Winjum <i>et al.</i> 1998 give the following proportions for wood products with short-term (<5 yr) uses after which they are retired and oxidized (applicable internationally): <table> <tr> <td>Sawnwood</td><td>0.2</td></tr> <tr> <td>Woodbase panels</td><td>0.1</td></tr> <tr> <td>Other industrial roundwood</td><td>0.3</td></tr> <tr> <td>Paper and Paperboard</td><td>0.4</td></tr> </table> The methodology makes the assumption that all other classes of wood products, and where wood product class ty is unknown, are 100% oxidized within 5 years. Therefore SLF, by wood product class, is equal to: <table> <tr> <th>Wood Product Class</th><th>SLF</th></tr> <tr> <td>Sawnwood</td><td>0.2</td></tr> <tr> <td>Woodbase panels</td><td>0.1</td></tr> <tr> <td>Other industrial roundwood</td><td>0.3</td></tr> <tr> <td>Paper and paperboard</td><td>0.4</td></tr> <tr> <td>Other classes of wood products</td><td>1.0</td></tr> </table>	Sawnwood	0.2	Woodbase panels	0.1	Other industrial roundwood	0.3	Paper and Paperboard	0.4	Wood Product Class	SLF	Sawnwood	0.2	Woodbase panels	0.1	Other industrial roundwood	0.3	Paper and paperboard	0.4	Other classes of wood products	1.0
Sawnwood	0.2																				
Woodbase panels	0.1																				
Other industrial roundwood	0.3																				
Paper and Paperboard	0.4																				
Wood Product Class	SLF																				
Sawnwood	0.2																				
Woodbase panels	0.1																				
Other industrial roundwood	0.3																				
Paper and paperboard	0.4																				
Other classes of wood products	1.0																				
Source of data:	The source of data is the published paper of Winjum <i>et al.</i> 1998 ¹																				
Justification of choice of data or description of measurement methods and procedures applied:	-																				
Any comment:	Parameter values to be updated if new empirically-based peer-reviewed findings become available.																				

¹ Winjum, J.K., Brown, S. and Schlamadinger, B. 1998. Forest harvests and wood products: sources and sinks of atmospheric carbon dioxide. *Forest Science* 44: 272-284

Data Unit / Parameter:	WW_{ty}
Data unit:	Dimensionless
Used in equations:	2, 4
Description:	WW = Fraction of extracted biomass effectively emitted to the atmosphere during production by class of wood product ty Winjum <i>et al.</i> 1998 indicate that the proportion of extracted biomass that is oxidized (burning or decaying) from the production of commodities to be equal to 19% for developed countries, 24% for developing countries. WW is therefore equal to $C_{XB,ty}$ multiplied by 0.19 for developed countries and 0.24 for developing countries.
Source of data:	The source of data is the published paper of Winjum <i>et al.</i> 1998 ²
Justification of choice of data or description of measurement methods and procedures applied:	-
Any comment:	Parameter values to be updated if new empirically-based peer-reviewed findings become available.

6.2. Data and Parameters Monitored

Data Unit / Parameter:	A_i
Data unit:	ha
Used in equations:	3
Description:	Total area of stratum i

Source of data:	The source of data shall be chosen with priority from higher to lower preference as follows: (a) Direct forest inventory of the project area, distinguishing commercially viable stocks on the basis of species and tree size, referencing local expert knowledge or a participatory rural assessment (PRA) of harvest practices and markets; (b) Forest inventory from a proxy area in the same region, representing the same forest type and age class, distinguishing commercially viable stocks on the basis of species and tree size, referencing local expert knowledge of harvest practices and markets National and forest type-specific or eco-region-specific (e.g. from National GHG inventory).
Description of measurement methods and procedures to be applied:	This parameter is updated at baseline renewal when aboveground biomass is re-inventoried as per module CP-AB (at least every 10 years).
Frequency of monitoring/recording:	At a minimum every time the baseline is updated (at least every 10 years).
QA/QC procedures to be applied:	
Any comment:	<i>Ex-ante</i> it shall be assumed that strata area will remain constant.

Data Unit / Parameter:	$V_{ex,i}$
Data unit:	m ³
Used in equations:	3
Description:	The volume of timber in m ³ extracted from within the stratum (does not include slash left onsite), reported by wood product class and preferably species.
Source of data:	Timber harvest records and/or estimates derived from field measurements or remote assessments with aerial photography or satellite imagery.
Description of measurement methods and procedures to be applied:	
Frequency of monitoring/recording:	At a minimum every time the baseline is updated (at least every 10 years).
QA/QC procedures to be applied:	

Any comment:	Note that this volume does not include logging slash left onsite. Data compilers must also make sure that extracted volumes reported are gross volumes removed (i.e. reported volume does not already discount for estimated wood waste, as is often the practice in harvest records). Assignment of volume extracted to wood product class(es), be substantiated on the basis of participatory rural appraisal (PRA) findings (also used to assess potential for degradation in module M-MON) or records of timber sales. Assignment of volume extracted to species, must be substantiated on the basis of either PRA findings, harvest records, or a commercial inventory. Baseline removals will be known ex-ante. With project removals are classed as project emissions and where expected shall be detailed ex-ante alongside evidence on expected harvested volumes.
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DOCUMENT HISTORY

Version	Date	Comment
v1.0	3 Dec 2010	Initial version released
v1.1	20 Nov 2012	The module was updated to appropriately account for the decay of carbon harvested wood products pool and the following revisions were made: • Equations 2 and 5 are revised so that only the wood waste fraction is immediately released. • Equations 3 and 6 are revised to calculate the total amount of wood that is expected to be emitted over a 100-year timeframe which include both the short-term and medium-term portions.